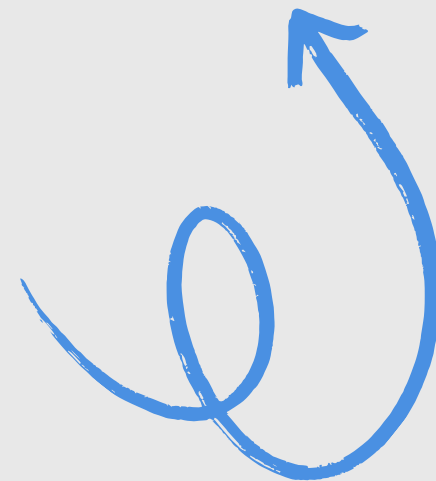


ReadyLine Predictive Maintenance Copilot

Team Nakshatra - IBM Bob AI
Hackathon



The Problem

Unreliable Asset Readiness Tracking

Military organizations struggle with accurate mission readiness assessments, relying on fixed maintenance schedules rather than analyzing real-time HUMS sensor data, leading to unexpected failures and costly downtime.



Why This Matters

Addressing an Ongoing DoD Challenge

The Department of Defense faces a critical, **long-standing issue** with maintenance practices that fail to ensure asset readiness, resulting in costly inefficiencies and operational risks.



Predictive Maintenance Copilot

A Solution for Asset Readiness

Informed Decision-Making

The copilot processes sensor data, providing clear insights into asset readiness, enabling maintenance teams to make informed decisions based on real-time conditions rather than fixed schedules.

Predictive Insights

By analyzing historical and live data, the system predicts potential failures ahead of mission timelines, helping to prevent unexpected downtimes and optimizing maintenance schedules effectively.

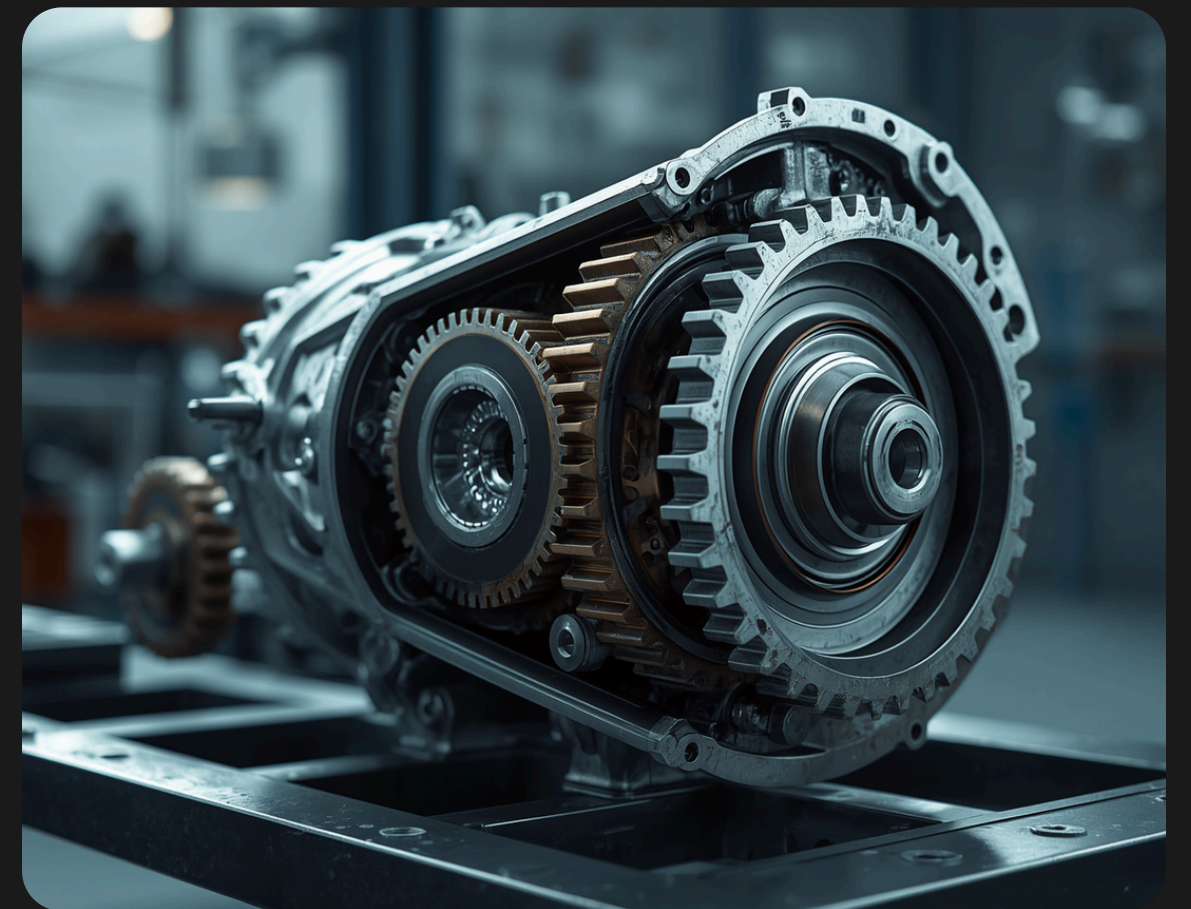
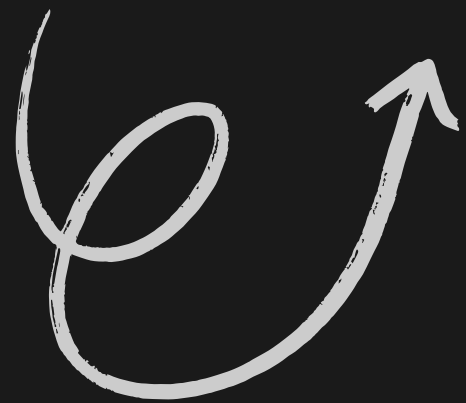
Prioritized Maintenance

The copilot recommends maintenance actions based on urgency and parts availability, facilitating a proactive approach that enhances fleet readiness while minimizing unnecessary resource expenditures.



Dual Modality Fleet Coverage

Engine and Gearbox Condition Insights



How It Works

Five Stages of Data Processing

Data Integration

Our copilot merges sensor, service, and parts data into a unified asset schema, ensuring comprehensive visibility of each asset's condition and readiness status for informed decision-making.

Predictive Analysis

Utilizing XGBoost and IBM Granite Time Series TTM, we forecast potential failures and estimate remaining useful life, enabling proactive maintenance planning and reducing unexpected asset downtime.

Readiness Explanations

Local IBM Granite via Ollama generates clear, plain language explanations of readiness issues, citing multiple corroborating sensors, fostering user trust and understanding of the system's assessments.



IBM Technology Integration

Core Components for Enhanced Readiness

Integrated Framework

Our solution leverages **IBM Bob** and **IBM Granite** for seamless data integration, ensuring a robust framework that supports predictive maintenance and asset readiness assessments across military operations.

Data Compatibility

We identified and resolved compatibility issues in the **IBM Granite Time Series TTM**, optimizing it for our specific data requirements, leading to more accurate failure predictions and reliable readiness insights.

Local Deployment

The **IBM Granite Ollama** system is designed for disconnected environments, ensuring maintainers can access critical insights and explanations without reliance on continuous network connectivity, fostering operational resilience.



Built for Trust

Design Focused on Real-World Adoption

Trust Building

Our system prioritizes **reliable explanations** to ensure maintainers trust the data, requiring multiple corroborating sensors before flagging readiness issues to minimize false alarms.

Prioritization Logic

We factor in parts lead times alongside urgency, ensuring a balanced approach to maintenance that aligns with **real-world operational** constraints and enhances trust in the system.



Acknowledging Limitations

Understanding Our Prototype's Scope and Constraints

Our solution employs XGBoost and TTM features instead of a comprehensive forecasting pipeline, and relies on synthetic data, highlighting the need for further development and real-world integration.



Future Developments

HUMS Data Integration

Integrating real HUMS data for specific platforms will enhance predictive capabilities, enabling more accurate assessments of asset readiness and operational efficiency for mission-critical scenarios.

Maintainer Feedback Loop

Establishing a feedback loop from maintainers aims to refine predictive algorithms, reducing false alarm rates while improving overall trust in the system's reliability and accuracy over time.

Full Integration

Achieving complete integration with watsonx Orchestrate will streamline operations, addressing provisioning constraints and enhancing the system's ability to deliver actionable insights for maintenance decision-making.

ReadyLine

**Driving Operational Efficiency
Forward**

